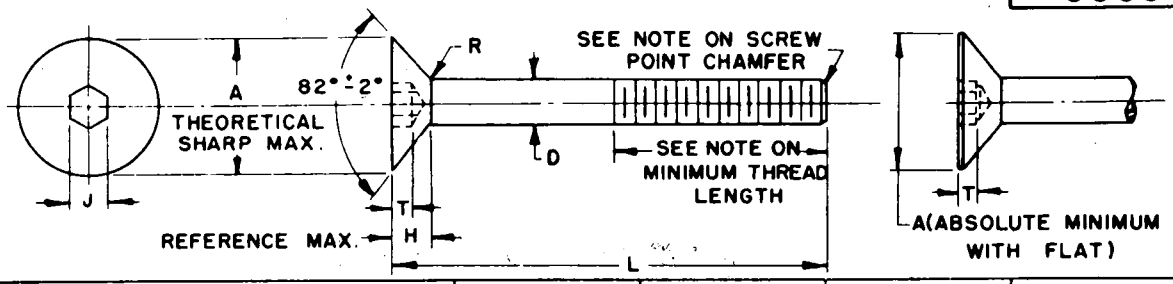
FED. SUP CLASS
5305



NOMINAL SIZE		#4 (.112)	#6 (.138)	#8 (.164)	#10 (.190)
THREADS PER INCH		40UNC-3A	32UNC-3A	32UNC-3A	24UNC-3A
D BODY DIAMETER	MAX	.112	.138	.164	.190
	MIN	.1096	.1353	.1613	.1867
A HEAD DIAMETER	THEO. SHARP MAX	.255	.307	.359	.411
	ABS. MIN	.218	.263	.311	.359
H HEAD HEIGHT	REFERENCE MAX	.083	.097	.112	.127
J SOCKET WIDTH ACROSS FLATS	MAX	.0635	.0791	.0947	.1270
	MIN	.0625	.0781	.0937	.1250
T KEY ENGAGEMENT	MIN	.055	.066	.076	.087
R RADIUS	MAX	.011	.014	.016	.019
* MINIMUM BREAKING STRENGTH	LBS	920	1400	2170	2710

L LENGTH	TOL.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
1/4		1	7		
3/8		2	8	13	19
1/2	+1/64	3	9	14	20
5/8	-1/64	4	10	15	21
3/4		5	11	16	22
1				17	23
1-1/4	+1/32				24
1-1/2	-1/64				25

(A) *BASED ON 160,000 PSI MIN TENSILE STRENGTH. BREAKING STRENGTHS ARE DETERMINED BY THE STRESS AREAS INDICATED IN SCREW-THREAD STANDARDS FOR FEDERAL SERVICES, HANDBOOK H-28.

(A) MATERIAL: STEEL, ALLOY, IN ACCORDANCE WITH FEDERAL STANDARD #66.

HARDNESS: ROCKWELL C36-43.

PROTECTIVE COATING: UNCOATED.

(A) THREADS: THE THREADS SHALL BE IN ACCORDANCE WITH SPECIFICATION MIL-S-7742.

MINIMUM THREAD LENGTH: THE MINIMUM USABLE THREAD LENGTH SHALL BE EQUAL TO 2D PLUS 1/2". THE THREAD LENGTH TOLERANCE SHALL BE EQUAL TO THE NOMINAL DIAMETER. SCREWS TOO SHORT FOR APPLICATION OF THE FORMULA SHALL BE THREADED AS CLOSE TO THE HEAD AS PRACTICABLE.

POINT CHAMFER: THE POINTS SHALL BE FLAT AND CHAMFERED. THE FLAT SHALL BE NORMAL TO THE AXIS OF THE SCREW. THE CHAMFER SHALL EXTEND SLIGHTLY BELOW THE ROOT OF THE THREAD AND THE EDGE BETWEEN THE FLAT AND THE CHAMFER MAY BE ROUNDED. THE INCLUDED ANGLE OF THE THREADED POINT SHALL BE APPROXIMATELY 90°. CHAMFERING OF SCREW SIZES UP TO AND INCLUDING #8 SHALL BE AT THE OPTION OF THE MANUFACTURER.

DIMENSIONS: ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.

PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS NUMBER, PLUS THE DASH NUMBER. EXAMPLE: MS24669-1

NOTES: (1) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
(2) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.

(A) FOR CHANGES SEE SHEETS 1 AND 2.

APPROVED GAPRGO REVISED (A)

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of the Navy and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A.	TITLE	
ORD Other Cust SHIPS	SCREW, CAP, SOCKET HEAD, FLAT COUNTERSUNK, ALLOY STEEL, UNCOATED, UNC-3A	MILITARY STANDARD MS24669
PROCUREMENT SPECIFICATION FF-S-86	SUPERSEDES:	SHEET 1 OF 2

DD FORM 672-1 (Coordinated) SEP 67 PREVIOUS EDITION OF THIS FORM MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 20240-23T14:59Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS 5305

NOMINAL SIZE		1/4	5/16	3/8	7/16
THREADS PER INCH		20UNC-3A	18UNC-3A	16UNC-3A	14UNC-3A
D BODY DIAMETER	MAX	.250	.3125	.375	.4375
	MIN	.2464	.3084	.3705	.4326
A HEAD DIAMETER	THEO. SHARP MAX	.531	.656	.781	.844
	ABS. MIN	.480	.600	.720	.781
H HEAD HEIGHT	REFERENCE MAX	.161	.198	.234	.234
J SOCKET WIDTH ACROSS FLATS	MAX	.1582	.1895	.2207	.2520
	MIN	.1562	.1875	.2187	.2500
T KEY ENGAGEMENT	MIN	.111	.135	.159	.159
R RADIUS	MAX	.025	.031	.037	.044
*MINIMUM BREAKING STRENGTH (A)	LBS	4950	8190	12100	16700

L LENGTH	TOL	DASH NO.	DASH NO.	DASH NO.	DASH NO.
3/8, 1/2, 5/8, 3/4, 1	+1/64 -1/64	27, 28, 29, 30, 31	37, 38, 39, 40, 41	49, 50, 51, 52	62, 63
1-1/4, 1-1/2, 1-3/4, 2	+1/32 -1/64	32, 33, 34, 35	42, 43, 44, 45	53, 54, 55, 56	64, 65, 66, 67
2-1/4, 2-1/2, 2-3/4, 3	+1/32 -1/32		46, 47	57, 58, 59, 60	68, 69, 70, 71

NOMINAL SIZE		1/2	5/8	3/4	
THREADS PER INCH		13UNC-3A	11UNC-3A	10UNC-3A	
D BODY DIAMETER	MAX	.500	.625	.750	
	MIN	.4948	.6191	.7436	
A HEAD DIAMETER	THEO. SHARP MAX	.937	1.188	1.438	
	ABS. MIN	.872	1.112	1.355	
H HEAD HEIGHT	REFERENCE MAX	.251	.324	.396	
J SOCKET WIDTH ACROSS FLATS	MAX	.3155	.3780	.3780	
	MIN	.3125	.3750	.3750	
T KEY ENGAGEMENT	MIN	.172	.220	.270	
R RADIUS	MAX	.050	.050	.050	
*MINIMUM BREAKING STRENGTH (A)	LBS	22300	35600	52800	

L LENGTH	TOL	DASH NO.	DASH NO.	DASH NO.	DASH NO.
3/4, 1	+1/64 -1/64	73, 74			
1-1/4, 1-1/2, 1-3/4, 2	+1/32 -1/64	75, 76, 77, 78	84, 85, 86, 87	93, 94, 95, 96	
2-1/4, 2-1/2, 2-3/4, 3	+1/32 -1/32	79, 80, 81, 82	88, 89, 90, 91	97, 98, 99, 100	

*NOTE: FOR ILLUSTRATION, MATERIAL, PROTECTIVE COATING AND OTHER PERTINENT DATA, SEE SHEET 1.

APPROVED GAPRGO REVISED. (A) FOR CHANGES SEE SHEETS 1 AND 2.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. ORD
Other Cust
SHIPS

TITLE
SCREW, CAP, SOCKET HEAD, FLAT COUNTERSUNK, ALLOY STEEL, UNCOATED, UNC-3A

MILITARY STANDARD
MS24669

PROCUREMENT SPECIFICATION
FF-S-86

SUPERSEDES:

SHEET
2

DD FORM 672-1 (Cancelled)

Source: https://assist.dla.mil -- Downloaded: 2026-0-23114:59Z
Check the source to verify that this is the current version before use.